

**THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS**

MSBA 7025 – Digital Experimentation Methods

GENERAL INFORMATION			
Instructor: Prof. Jing Ouyang Email: jingoy@hku.hk Office: KK Leung Building 1020 Phone: 3910-3107 Consultation times: By appointment Tutor: TBD Pre-requisite: MSBA 7002 Business Statistics Co-requisites: None			
COURSE DESCRIPTION			
The newly emerging capability to rapidly deploy and iterate online controlled experiments to assist decision makings in organizations is one of the most significant innovations in today's technology industry. As more and more social interactions, decisions, opinions, and transactions are mediated by online platforms, digital experiments are becoming increasingly crucial for firms to understand their user behaviors and make product decisions. This course will cover the most cutting-edge digital experimentation methods used in daily operations at large technology firms, such as Tencent, Alibaba, Facebook, and LinkedIn. We will also share the key lessons and pitfalls encountered in practice. Topics include the statistics behind experiments, experimental design, methods of analyzing experiments in a trustworthy way, advanced topics including sequential testing and multi-armed bandit, and recent development in digital experimentation. Students will also learn how to conduct and analyze online experiments using programming languages, such as python, in assignments and a course project.			
COURSE OBJECTIVES			
1. Obtain solid understanding of the principles and implementations of A/B testing 2. Formulate the business problem and develop statistical analysis skills to draw scientific conclusions 3. Implement the analytic methods using software tools 4. Present analysis results in business relevant language			
FACULTY LEARNING GOALS (FLGs)			
FLG1: Acquisition and internalization of knowledge of the programme discipline FLG2: Application and integration of knowledge FLG3: Inculcating professionalism FLG4: Developing global outlook FLG5: Mastering communication skills FLG6: Cultivating leadership			
COURSE LEARNING OUTCOMES			
Course Learning Outcomes		Aligned Faculty Learning Goals	
CLO1: Clearly identify and formulate a structured business problem		FLG1	
CLO2: Select and use effective method to address the business problem		FLG2	
CLO3: Use software tools to provide solution to the issue at hand		FLG2, 4	
CLO4: Communicate the solution effectively		FLG3, 5, 6	
COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		Expected contact hour	Study Load (% of study)

T&L1. Interactive lectures		30	25%
T&L2. Tutorials		8	6.7%
T&L3. Self-study		42	35%
T&L4. Group projects and individual assignments		40	33.3%
Total		120	100%
Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Participation	Attendance & discussions	10%	CLO1, 2, 3, 4
A2. Assignments	Effort and accuracy	40%	CLO1, 2, 3, 4
A3. Final project	Effort and accuracy	50%	CLO1, 2, 3, 4
Total		100%	
STANDARDS FOR ASSESSMENT			
Course Grade Descriptors			
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent performance in assignments		
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills • Good performance in assignments		
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable performance in assignments		
D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable performance in assignments		
F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills • Poorly performance in assignments		
Assessment Rubrics for Participation			

A+, A, A-	• High participation in discussions • Always attend in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Sometimes attend in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend in-class discussions • Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments and the Exam	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills
Assessment Rubrics for Project Report	
A+, A, A-	Provides persuasive original, insightful and well-reasoned recommendations that clearly follow from the analysis and effectively address the key issue
B+, B, B-	Provides well-reasoned recommendations that follow from the analysis and address the key issue
C+, C, C-	Conducts adequate analysis and provides reasoned recommendations that address the key issue

D+, D	Offers weak recommendations, which do not address key issue
F	Fails to identify key issue and/or provide adequate analysis and/or recommendations
Assessment Rubrics for Project Presentation	
A+, A, A-	Delivers a persuasive, engaging and impactful presentation, and provides well- reasoned answers to address all questions raised
B+, B, B-	Delivers an engaging presentation and provides well-reasoned answers to address all questions raised
C+, C, C-	Delivers an adequate presentation and provides answers to address questions raised
D+, D	Weak presentation and non-persuasive answers to questions raised
F	Failure to present appropriately and/or provide adequate answers to questions raised
COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE	
Session 1: Introduction to Online Controlled Experiments Sessions 2 – 3: Statistical Foundations • Basic statistical concepts • Common statistical tests • Understanding Type I and Type II errors Session 4: Design and Implement Experiments • Randomization techniques • Randomization unit and analysis unit • Metrics Session 5: Experiment Analysis • Variance estimation • Delta Method • A/A Tests Session 6: Internal and External Validity • Violation of SUTVA • Survivorship bias • Sample ratio mismatch • Heterogeneous treatment effect (Subgroup analysis, regression-based analysis) Session 7: Improve Sensitivity • Triggering • Stratification & Post stratification • CUPED Session 8: Advanced Topics • Multi-armed bandits Sessions 9: Industry application and Recent developments	

Session 10: Observational Causal Inference

RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

Kohavi, Ron, Diane Tang, and Ya Xu. *Trustworthy online controlled experiments: A practical guide to a/b testing*. Cambridge University Press, 2020.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

- Conducting mid-term survey in additional to SETL around the end of the semester
- Email communication

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

1. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Academic dishonesty also includes, but is not limited to, providing false or misleading information to receive a postponement or an extension on an exam or other assignment.
2. An orderly learning environment is extremely important for this course. Disruptive behaviors are inconsiderate to other students as well as to the instructor, and are absolutely unacceptable. Talking during lectures, arriving to class late, and any other disruptions of mobile devices are not allowed; students who are responsible for any of these actions will be subject to academic penalty and will be asked to leave the classroom.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

1. Lecture notes and self-learning materials will be uploaded on Moodle.
2. No late assignment submission will be accepted.
3. The instructor reserves all the rights to make necessary changes to the syllabus. If so, the changes will be announced as soon as possible.